

DRP Writeup

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Introduction

This is the writeup for the Winter 2026 DRP focusing on the Paper titled "Asymptotics of Maximum Likelihood without the LLN or CLT or Sample Size Going to Infinity" by Charles J. Geyer. The paper mainly focuses on deriving asymptotic results of Maximum Likelihood without assuming sample sizes going to infinity. This includes results like MLE estimators being asymptotically quadratic, Large Sample Distribution of MLEs, so on and so forth. Due to the nature of this Paper being more theoretically oriented and proof-heavy, this specific project is thus devoted more heavily to reading and comprehension, as opposed to applications of research. Thus, the write-up is focusing more on the summarization of the readings, as well as the discussions in meetings between the mentor and the mentee.

Preliminaries

Convergence in Distribution, Helly-Brays

Much of the rigor encountered in comprehension of this specific paper should begin with a discussion on introducing a notion of convergence for random variables. This may seem like an extension of the convergence of sequences and series in Undergraduate-level Real Analysis, yet a rigorous definition and a more ubiquitous definition of it based on expectation is presented below for the sake of formality.

Convergence in Distribution

Definition 1 (Convergence in Distribution). A sequence of random variables $X_1, X_2, \dots$, with X_n having distribution function F_n converges in distribution to a random variable X with distribution function F if

$$F_n(x) \rightarrow F(x), \quad \text{as } n \rightarrow \infty$$

for every real number x that is a continuity point of F . We indicate this by writing

$$X_n \xrightarrow{D} X, \quad \text{as } n \rightarrow \infty.$$

Translating this into more natural language, this is stating that if for all real numbers x that are continuity points of some function F , then if some sequence of distribution functions $F_n(X)$ associated with a sequence of random variables $X_1, X_2, \dots, X_n$ converges to the distribution function $F(X)$ associated with the random variable X , then that sequence of random variables $X_1, \dots, X_n$ itself converges in distribution to the random variable X .

Helly-Brays

Definition 2. [Helly-Bray] A sequence of random variables $X_1, X_2, \dots$ converges in distribution to a random variable X if and only if

$$E\{g(X_n)\} \rightarrow E\{g(X)\}$$

for every bounded continuous function $g: \mathbb{R} \rightarrow \mathbb{R}$.

Convergence in Probability

Definition 3 (Convergence in Probability to a Constant). A sequence of random variables $X_1, X_2, \dots$ converges in probability to a constant a if for every $\epsilon > 0$

$$P(|X_n - a| > \epsilon) \rightarrow 0, \quad \text{as } n \rightarrow \infty.$$

We indicate X_n converging in probability to a by writing

$$X_n \xrightarrow{P} a, \quad \text{as } n \rightarrow \infty.$$

Notice how this is just a special case of convergence of distribution, albeit converging to a constant rather than a distribution.

However, these are more of theoretical definitions than how convergence of random variables is shown in statistics. Rather, the Law of Large Numbers and the Central Limit Theorem are often used to show convergence results.

Central Limit Theorem

Theorem 1 (The Central Limit Theorem (CLT)). If $X_1, X_2, \dots$ is a sequence of independent, identically distributed random variables having mean μ and variance σ^2 and

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

is the sample mean for sample size n , then

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{D} Y, \quad \text{as } n \rightarrow \infty,$$

where $Y \sim \mathcal{N}(0, \sigma^2)$.

At first glance, this might look very different from the informal definition that may look something like this:

Theorem 2. Given any sequence of independent and identically distributed random variables $X_1, \dots, X_n$ and its sample means $\bar{X}_i$, as $n \rightarrow \infty$, $\bar{X}_i \approx \mathcal{N}(0, \sigma^2)$

Notice how that by the fact of the closure under convolution of Normal distributions, the algebraic expression that defines the Central Limit Theorem can be easily obtained by algebra (i.e. linearity of mean and properties of variance)

Law of Large Numbers

One place convergence in probability appears is in the law of large numbers.

Theorem 3 (Law of Large Numbers (LLN)). If $X_1, X_2, \dots$ is a sequence of independent, identically distributed random variables having mean μ , and

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

is the sample mean for sample size n , then

$$\bar{X}_n \xrightarrow{P} \mu, \quad \text{as } n \rightarrow \infty.$$

The only requirement is that the mean μ exist. No other property of the distribution of the X_i matters.

Giving a brief outline of these foundations of asymptotic theories in statistics also serves to motivate this research paper, as well as our project this quarter.

Maximum Likelihood Estimation

Likelihood Function, log-likelihood function

The Likelihood Function may seem strange for people accustomed to the elementary definitions of probability densities and cumulative densities. We will begin with the definition first.

Definition 4. Recall that the joint PMF/PDF of a random sample of size n is defined as $f(x_1, \dots, x_n)$. If we rewrite this definition to make the dependence on a parameter θ explicit, we obtain $f(x_1, \dots, x_n | \theta)$.

Normally, we treat the parameter θ as a constant and $(x_1, \dots, x_n)$ as variables. However, if we instead regard θ as the variable and $(x_1, \dots, x_n)$ as fixed constants, we obtain the likelihood function

$$L(\theta) = f(x_1, \dots, x_n | \theta),$$

that is, the joint PMF/PDF viewed as a function of the parameter.

The log likelihood $\ell(\theta)$ is simply defined by taking the log of the likelihood function. This is done mostly to simplify calculations, as the logarithmic function would preserve the maximum as well as transform multiplications into summations.

Maximum Likelihood Estimator

we can then write the likelihood function as $\mathcal{L}(\theta) = f_\theta(x_1, \dots, x_n)$. Our Maximum Likelihood Estimator $\hat{\theta}$ is then the value of θ that maximizes the likelihood function $\mathcal{L}(\theta)$. For simplicity in calculations, we usually take its logarithms and maximizes the resulting log-likelihood function $\ell(\theta) = \log(\mathcal{L}(\theta))$ instead. ML estimators have nice limiting properties.

- consistent estimator
- asymptotic unbiasedness, the estimator's bias goes to zero as sample size approaches infinity
- efficient, achieves Cramer-Rao lower bound when sample size approaches infinity
- invariance, if $\hat{\theta}$ is the MLE for θ , then $g(\hat{\theta})$ is the MLE for $g(\theta)$

Fisher Information, Cramer-Rao Lower Bound

Fisher Information

Let $f(x | \theta)$ be the pdf of X for all $\theta \in \Theta$, and let $\lambda(\theta) = \log f(X | \theta)$. We define its first and second partial derivatives with respect to θ as $\frac{\partial \lambda}{\partial \theta}$ and $\frac{\partial^2 \lambda}{\partial \theta^2}$. The first-order partial derivative

$$\frac{\partial}{\partial \theta} \log f(X | \theta)$$

is called the score.

Now, let X be a continuous random variable. Since

$$\int f(x | \theta) dx = 1,$$

differentiating both sides with respect to θ gives

$$\begin{aligned} \frac{\partial}{\partial \theta} \int f(x | \theta) dx &= \int \frac{\partial}{\partial \theta} f(x | \theta) dx \\ &= 0. \end{aligned}$$

Using $\frac{\partial}{\partial \theta} \log f(x | \theta) = \frac{1}{f(x | \theta)} \frac{\partial}{\partial \theta} f(x | \theta)$, we obtain

$$\mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X | \theta) \right] = 0,$$

so the score has mean 0.

The Fisher information is the variance of the score. In the continuous case, it is

$$I(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X | \theta) \right)^2 \right] - \mathbb{E}_\theta \left[\frac{\partial}{\partial \theta} \log f(X | \theta) \right]^2 = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X | \theta) \right)^2 \right] = \int_{\mathcal{X}} \left(\frac{\partial}{\partial \theta} \log f(x | \theta) \right)^2 f(x | \theta) dx,$$

and in the discrete case

$$I(\theta) = \sum_{x \in \mathcal{X}} \left(\frac{\partial}{\partial \theta} \log f(x | \theta) \right)^2 f(x | \theta).$$

Equivalently, under regularity conditions,

$$I(\theta) = -\mathbb{E}_\theta \left[\frac{\partial^2}{\partial \theta^2} \log f(X | \theta) \right].$$

For an i.i.d. sample $X_1, \dots, X_n$ with joint density $f(x^n | \theta) = \prod_{i=1}^n f(x_i | \theta)$, the total Fisher information is

$$I_n(\theta) = nI(\theta),$$

and $I(\theta)$ is called the Fisher information per observation.

Intuitively, this Fisher information measures how much a test statistic T gives information about the parameter θ by weighting each potential outcome θ with the corresponding pmf/pdf $f(x|\theta)$ (Overall sensitivity to changes). From the definition involving the expectation of the second partial derivatives, this could be also understood intuitively as the measure of sensitivity or "curvature" of the log-likelihood function around the parameter

Cramer-Rao Lower Bound

The Cramér–Rao Lower Bound answers the question: how small can the variance of an unbiased estimator be? Formally:

Theorem 4. Let $\hat{\theta}_n$, $n \in \mathbb{N}$, be an unbiased estimator of $g(\theta)$ based on an i.i.d. sample of size n with density $f(x | \theta)$. Then, under regularity conditions,

$$\text{Var}_\theta[\hat{\theta}_n] \geq \frac{[g'(\theta)]^2}{n I(\theta)},$$

where

$$I(\theta) = \mathbb{E}_\theta \left[\left(\frac{\partial}{\partial \theta} \log f(X | \theta) \right)^2 \right]$$

is the Fisher information for θ .

These two quantities are useful because they are the parameters of the large sample distribution of MLE that we've mentioned before:

Theorem 5. Suppose $X_1, X_2, \dots, X_n \sim f_\theta(x)$ indexed by a parameter θ , with true but unknown value θ_0 . Then, for large values of n , under certain regularity conditions, θ_0^{mie} is approximately normally distributed with mean θ_0 and standard error $\frac{1}{\sqrt{I(\theta_0)}}$, where $I(\theta_0) = -E\left[\frac{d}{d\theta^2} \ell(\theta)\right]_{\theta=\theta_0}$

Chapter 1

1.1

If the log likelihood function is approximately of the form $l(\theta) = U + Z'\theta - \frac{1}{2}\theta'K\theta$ with K independent of the parameter θ (invariant in law), then the maximum likelihood estimate $\hat{\theta} \sim N(\theta, K^{-1})$.

Notice how the form $l(\theta) = U + Z'\theta - \frac{1}{2}\theta'K\theta$ is a generalization or matricized form of a second degree Taylor Polynomial expansion:

$$l(\theta) \approx l(\hat{\theta}) + (\theta - \hat{\theta})\nabla f(\hat{\theta}) - \frac{1}{2}(\theta - \hat{\theta})\nabla^2 l(\hat{\theta})(\theta - \hat{\theta})$$

To see its asymptotic behaviors,

1.2

The argument for investigating Asymptotics without sample size n approaching infinity: mathematically, the convergence of a sequence gives no information about the initial segment of the sequence. A similar argument can be made here in a statistical context. The author points out that the assumption of sample size approaching infinity is absurd and sometimes redundant, in that it does not distinguish the specific estimators that are being used to approximate the MLE asymptotically. The author further delineates his approach towards the "no n asymptotics" approach. Here, no n refers to using n as an index, but never to represent the sequence of statistics with sample size n . In addition, both the epsilon-delta definition, and the sequence definition of convergence can be used.

1.3

The use of asymptotics creates a paradoxical loop: in reality, asymptotic theories are heuristics. Therefore, we will use bootstrap methods to approximate results. However, the only justification of bootstrap methods (both parametric and non-parametric) are asymptotics.

Chapter 2

2.1

The log likelihood of a parametric family of probability distributions with density $f_\theta, \theta \in \Theta$ with respect to a measure λ is given as $l(\theta) = u(x) + \log f_\theta(X), \theta \in \Theta$. As we are only interested in almost surely positive densities $f_\theta(x)$, all arguments of $\log f_\theta(x)$ are never zero, thus $l(\theta)$ is always well-defined. Then, we have the following: $E_{\theta+\delta}\{g(x)\} = \int g(x)f_{\theta+\delta}(x)\lambda dx = \int g(x)\frac{f_{\theta+\delta}(x)}{f_\theta(x)}f_\theta(x)\lambda dx = \int g(x)e^{l(\theta+\delta)-l(\theta)}f_\theta(x)\lambda dx = E_\theta\{g(x)e^{l(\theta+\delta)-l(\theta)}f_\theta\}$

2.2

Suppose the log likelihood $l(\theta)$ is defined as asymptotically quadratic so that $l(\theta) = U + Z^T\theta - \frac{1}{2}\theta^TK\theta$. Since the constant term U is analogous to the term $u(X)$ in $l(\theta) = u(x) + \log f_\theta(X)$, we are thus mainly interested in the log likelihood ratios $l(\theta + \delta) = (Z - K\theta)^T\delta - \frac{1}{2}\delta^TK\delta$.

Then we have the following theorem (LAMN): Suppose $l(\theta) = U + Z^T\theta - \frac{1}{2}\theta^TK\theta$ is the log likelihood of a probability model, then :

- K is almost surely positive semi-definite

Also, these two following conditions are equivalent:

- the conditional distribution of Z given K for parameter value θ is $N(K\theta, K)$
- the distribution of K does not depend on the parameter θ

2.3

Other Tools for dealing with Asymptotics

- Delta Method
- Continuous Mapping Theorem
- Portemanteau theorem
- Prokhorov's theorem

References

1. Convergence in distribution:

- C. J. Geyer, *Stat 5101 Lecture Notes*. Available at <https://www.stat.umn.edu/geyer/5101/notes/n1.pdf>. [file:1]
- R. Lockhart, *STAT 801: Mathematical Statistics – Convergence in Distribution*. Available at https://www.sfu.ca/~lockhart/richard/801/04_1/lectures/conv_in_dist/web.pdf. [file:2]

2. Maximum likelihood estimation:

- S. Han, *Lecture 11: Maximum Likelihood Estimators*. Available at <https://shaobohan.net/sta111/Lec11.pdf>. [web:23]